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Original Research

ICD2Vec: Mathematical representation of diseases

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ABSTRACT

Background: The International Classification of Diseases (ICD) codes represent the global standard for reporting disease conditions. The current ICD codes connote direct human-defined relationships among diseases in a hierarchical tree structure. Representing the ICD codes as mathematical vectors helps to capture nonlinear relationships in medical ontologies across diseases.

Methods: We propose a universally applicable framework called "ICD2Vec" designed to provide mathematical representations of diseases by encoding corresponding information. First, we present the arithmetical and semantic relationships between diseases by mapping composite vectors for symptoms or diseases to the most similar ICD codes. Second, we investigated the validity of ICD2Vec by comparing the biological relationships and cosine similarities among the vectorized ICD codes. Third, we propose a new risk score called IRIS, derived from ICD2Vec, and demonstrate its clinical utility with large cohorts from the UK and South Korea.

Results: Semantic compositionality was qualitatively confirmed between descriptions of symptoms and ICD2Vec. For example, the diseases most similar to COVID-19 were found to be the common cold (ICD-10: J00), unspecified viral hemorrhagic fever (ICD-10: A99), and smallpox (ICD-10: B03). We show the significant associations between the cosine similarities derived from ICD2Vec and the biological relationships using disease-to-disease pairs. Furthermore, we observed significant adjusted hazard ratios (HR) and area under the receiver operating characteristics (AUROC) between IRIS and risks for eight diseases. For instance, the higher IRIS for coronary artery disease (CAD) can be the higher probability for the incidence of CAD (HR: 2.15 [95% CI 2.02–2.28] and AUROC: 0.587 [95% CI 0.583–0.591]). We identified individuals at substantially increased risk of CAD using IRIS and 10-year atherosclerotic cardiovascular disease risk (adjusted HR: 4.26 [95% CI 3.59–5.05]).

Conclusions: ICD2Vec, a proposed universal framework for converting qualitatively measured ICD codes into quantitative vectors containing semantic relationships between diseases, exhibited a significant correlation with actual biological significance. In addition, the IRIS was a significant predictor of major diseases in a prospective study using two large-scale datasets. Based on this clinical validity and utility evidence, we suggest that publicly available ICD2Vec can be used in diverse research and clinical practices and has important clinical implications.

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1. Introduction

Statement of significance.

Issue.
It is important to represent clinical concepts as embedding vectors that can be applicable to various fields of biomedical and translational research.

What is Already Known.
Previous embedding techniques for medical concepts were used to calculate the similarity of phenotypes, predict diseases, reduce dimensions, or identify clinical terms. However, no studies have validated the embedded medical vectors by comparing them with biological estimates and evaluating the model in multiple datasets of diverse ancestry.

What this Paper Adds.
We developed a universal framework to convert ICD codes into mathematical vectors that capture semantic relationships among diseases. We evaluated our algorithm in terms of biological factors. We further suggested an individual risk score and validated the score using two large datasets.

1.1. Background

In the past few decades, hospitals worldwide have rapidly adopted electronic health records (EHRs). Recently, more than 84% and 94% of hospitals have adopted EHRs in the UK and US, respectively [1,2]. Standardized coding systems are the basis for rapid advances in healthcare systems. The tenth revision of the International Classification of Diseases (ICD-10) is a global standard for health data, clinical documentation, and statistical aggregation [3,4]. It defines categorical codes for diseases, signs and symptoms, abnormal findings, complaints, social circumstances, and external causes of injuries or diseases [5–7]. These diagnosis codes are practical because they provide a common ground for the reporting and monitoring of diseases, allowing hospitals to share data in a consistent way. However, despite the complex relationships across diseases, current ICD codes only contain hierarchical classifications [4].

In addition to their importance in clinical practice, ICD codes are essential for biomedical research, e.g., to define the progress in disease conditions and as predictor and outcome variables for analyses of prospective disease risks [8–12]. In epidemiology, the variables for the prevalence of a disease are conventionally coded using a binary dummy coding representing the presence or absence of the disease, regardless of the implicit relationships between the predictor and outcome variables [13,14]. In machine learning, the widely used one-hot encoding can also miss such relationships between variables, which is called the curse of dimensionality, leading to a decrease in performance or slow convergence speed. In contrast, if the mathematically defined ICD variables contain the relational features between diseases, the performances of corresponding models may be better than simple one-hot encoding-based models. Efficient embedding is important for reducing the dimensionality of the input data while maintaining the intrinsic information, thus enabling the development of a lightweight model [15,16]. The contextual embedding of clinical concepts such as Med2Vec has been investigated in previous studies [17–20]. However, the representation learning using EHRs has sampling bias. Hence, an objective representation of ICD codes as mathematical vectors is necessary.

In this work, we proposed ICD2Vec, a framework for converting ICD codes into mathematical vectors (Fig. 1). We demonstrated that the semantic relationships were preserved between ICD codes by finding an ICD code acting as an arithmetical combination of multiple symptoms or ICD codes. In addition, to confirm the validity of the vectorized ICD codes, we compared the semantic relationships with the biological relationships between diseases generated from other sources [21]. Finally, to illustrate the clinical utility of ICD2Vec, we proposed an individual risk score derived from ICD2Vec for predicting the risk of subsequent diseases based on a history of diagnosed diseases of an individual relative to a baseline.

1.2. Related works

Natural language processing (NLP) algorithms have been developed based on innovative neural networks. In particular, with the advantages of neural networks, high-dimensional concepts can be represented with meaningful low-dimensional vectors. A skip-gram is an algorithm for vectorizing a word by predicting the adjacent words in a sentence [22]. It enables a semantic relationship between vectors, such as “king-queen” to “man-woman.” However, it cannot represent a completely new word. FastText [23] constructs a general model to train words in a sentence, and a subword model to additionally train a bag of characters in a word. It can estimate out-of-vocabulary words using subwords. Recently, the state-of-the-art results have been achieved by BERT with a new algorithm and numerous parameters [24]. BERT models proposed the contextual embeddings that learn the semantic meaning of a sentence. Interestingly, the larger-scale transformer models comprising billions of parameters achieved the higher scores on NLP tests [25,26]. We used various BERT-based models trained with English corpus and medical information.

These representation-learning methods are now very relevant in the medical field. In particular, medical concept embedding has been an active area of research. Previous studies have attempted to summarize sparse and high-dimensional medical concepts using compressed vectors [27,28]. Choi *et al.* transformed medical concepts into arithmetic vectors using a skip-gram from electrical health data [29]. A medical concept was trained by predicting based on the time-adjacent diagnostic records of a subject. The transformed medical concept vectors improved the performance of a heart-failure-risk prediction model. Minarro-Gimenez *et al.* applied a skip-gram to medical corpora collected from medical websites [30]. Bai *et al.* proposed a pipeline for calculating a modified pointwise mutual information score and used skip-grams with negative sampling from different medical ontologies [31]. BioBERT is a pre-trained representation model for biomedical texts [17]. The authors provided weights and source codes for the model. GatorTron [32] is a large clinical language model fine-tuned with Megatron [26] which contains large-scale parameters. In general, existing models have successfully learned the concept of embedding particular tasks at various levels, but they have not been able to effectively identify the similarities and causal interactions between all diagnosis codes.

2. Methods

2.1. Dataset

We used the 2019 ICD-10-CM, which is a clinical modification of ICD-10 in the United States and additionally extends some medical conditions or illnesses, relevant clinical information, and the approximate synonyms [33]. The disease names from the ICD codes were used to encode the information of each code from the Clinical Table Search Service provided by the National Institutes of Health (NIH) of the United States. Additionally, the clinical information and approximate synonyms of the ICD codes were incorporated into their vector representations. The clinical information was obtained from official medical websites, such as NIH and a summary in MedlinePlus provided by the Centers for Disease Control and Prevention. We extracted the comprehensive definitions and information for ICD-10 from a publicly available web database. Incorporating this information into the vector representation enabled us to obtain a representation of the disease codes (Fig. 1). In this study, we used 24,354 ICD codes consisting of categories and subcategories with up to 5 characters. The ICD codes contain 139,932 words and 540,747 words with clinical information and approximate synonyms. The information is contained with an average of 65.5 words for each ICD code, for example, 13,439 ICD codes represented with under 10 words (Table 1).

Our embedded vectors were evaluated for generalization using a large prospective cohort from the UK Biobank and massive EHR data

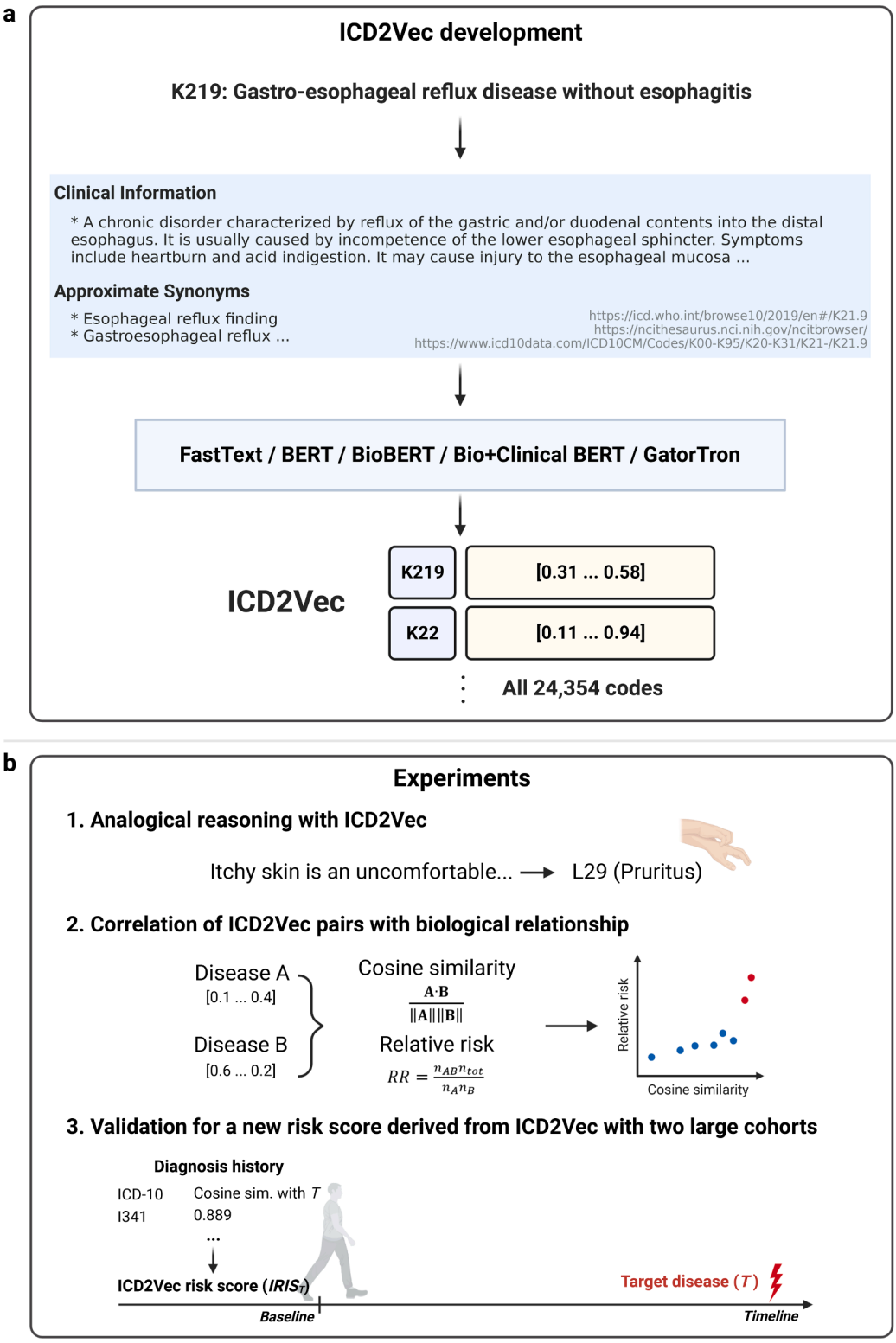

Fig. 1. ICD2Vec overview. (a). Illustration for encoding International Classification of Diseases (ICD) codes to vectors. We obtained disease definitions, clinical information, and approximate synonyms for all 24,354 ICD codes. For example, using GatorTron, every ICD code was converted to a corresponding 1,024-dimensional vector. Finally, we calculated an average for the word vectors to indicate an ICD code. (b) We validated the ICD2Vec with analogical reasoning using arithmetic operations and observed significant associations in the cosine similarities between pairs of ICD codes and their biological relationships. We suggested a new "ICD2Vec risk score" (IRIS) derived from ICD2Vec. The IRIS is a value representing the cosine similarities between the target disease and diagnosed diseases for a person in a specific time period. The individual risk for the target disease was measured using the IRIS according to the diagnosis history. We validated the clinical utility of the IRIS as a risk score with two datasets from the UK and South Korea. This figure was created with BioRender.com.

Table 1
Range of the number of words including information for each ICD code.

Range	[1 to 10]	[11 to 100]	[101 to 500]	greater than 500
Number of ICD codes	13,439	9,360	1,509	46

from the Clinical Data Warehouse (CDW) of the Samsung Medical Center (SMC). The UK Biobank provides approximately 500,000 participants' medical histories from interviews and questionnaires, as well as biological samples for genotyping and informed consent for long-term medical follow-up through links to national health registries. This research was conducted using the UK Biobank Resource under Application Number 33002. Second, the EHRs of over 200,000 patients who visited the SMC between 2010 and 2012 were collected in a hospital information system, i.e., the CDW of SMC. We could access and retrieve deidentified records, including demographic information and the diagnosis histories of the participants. Ethical approval for the study protocol was granted by the Institutional Review Board (No: 2021-07-016) at SMC.

2.2. ICD2Vec development

The word embeddings in NLP are a type of word representation in a high-dimensional space, such that words with similar meanings have similar representations and are positioned close to each other in the vector space. Word embeddings help the machine capture the context of human language. They are used to efficiently solve NLP problems owing to their ability to capture the semantic relationships among words.

ICD2Vec, which converts qualitative ICD to quantitative vectors, is a simple and flexible algorithm that easily combined with NLP algorithms. We developed ICD2Vec based on term frequency-inverse document frequency (TF-IDF), FastText, and four BERT algorithms and we compared these with existing algorithms such as Med2Vec [19] and ICD-Codex [34]. Detailed descriptions of Med2Vec, ICD-Codex, TF-IDF, FastText, and four BERT algorithms are described in **Supplementary Information**. We obtained the pre-trained BERT models from a unified API in Hugging Face Open-Source Libraries [35]. BERT was used with WordPiece embedding with a 30,000-token vocabulary, and pre-trained on a large English corpus [36]. BioBERT was pre-trained with PubMed abstracts and PubMed Central full-text articles [17]. Bio + Clinical BERT based on BioBERT was further pre-trained with clinical notes in the MIMIC-III v1.4 database [37]. GatorTron was trained with clinical notes of University of Florida Health, PubMed, Wikipedia, and MIMIC-III databases [32]. GatorTron-OG comprising 345 million parameters is publicly available using the NVIDIA NeMo framework. The paths for accessions of the models are described in **Data sharing**.

We fine-tuned these four models using masked language modeling (MLM) with our ICD-10 corpus. MLM is a self-supervised method that trains a model to predict a masked word in a given sentence. We randomly masked words of up to 30% in each ICD-10 description for fine-tuning. A description for each ICD code was configured with definitions, clinical information, and approximate synonyms of the ICD-10 code. We truncated the following words in a sentence over 500 words. We feed the tokenized words to the model, then extracted the features of the last layer in the model. Because we fine-tuned the words within sentences using MLM, we applied average pooling to these token vectors to obtain a summarized representation of the sentence. It simply generated the vectors, which is called ICD2Vec, for 768 dimensions for BERT and Bio + Clinical BERT, and 1,024 dimensions for BioBERT and GatorTron, respectively.

2.3. ICD2Vec risk score (IRIS)

The most advantageous feature of ICD2Vec is that the disease definition, as a high-dimensional concept, can be easily expressed as an arithmetic vector. Because the vectors have meaningful relationships with each other, it is possible to adapt the vectors as a clinical variable for epidemiological or association studies. We suggested a disease-wide risk score for quantitatively measuring the risk of a target disease using ICD2Vec.

The disease-wide risk score (R_i) for a diagnosed disease calculated using the cosine similarity (CS) between a vector for a disease ($\mathbf{D_i}$) and a vector for a target disease ($\mathbf{T}$) was defined as.

$$R_i = CS(\mathbf{D_i}, \mathbf{T}) = \frac{\mathbf{D_i} \bullet \mathbf{T}}{\|\mathbf{D_i}\|\|\mathbf{T}\|}$$

The IRIS was defined using the top three averages or using the maximum disease-wide risk scores in an individual. Fig. 3 illustrates the derivation of the IRIS using the disease-wide risk score from the past medical history. The past medical history, recorded with ICD-10, was obtained from the patients who were diagnosed with any disease at least once for a fixed duration before the start of the follow-up as the baseline. For example, assuming an individual is diagnosed with 10 diseases within the specific period, vectors ($\mathbf{D}_1 \cdots \mathbf{D}_{10}$) can be obtained from ICD2Vec using ICD codes for the diseases. IRIS_T for the target disease T can be defined as the average of the top three (R_{avg3}) or the maximum (R_{max}) disease-wide risk scores between $\mathbf{D}_1 \cdots \mathbf{D}_{10}$ and $\mathbf{T}$. Specifically, if we obtained the past medical history recorded with ICD-9, we mapped ICD-9 to ICD-10 using General Equivalence Mappings provided by the Centers for Medicare & Medicaid Services (https://www.cms.gov).

2.4. Study population for epidemiological studies

The UK Biobank. Of approximately 500,000 participants aged 40 to 69 years in the UK Biobank, we selected eligible patients for the epidemiological studies of CAD, T2D, dementia, and liver cancer, as follows. First, we defined the epidemiological baseline as a specific n year period (3, 5, 7, and 9 years) before the date of a participant's initial assessment. We excluded patients who were diagnosed with the definition of the target disease at the baseline (**Supplementary Table S4**). Additionally, we excluded participants who had missing values for the variables included in the analysis. We included participants who were diagnosed at least once to derive the IRIS. The date of onset of the target disease as the primary outcome was considered as the first date of diagnosis of the target disease. The participants were followed up until the earliest date among those of the event, the last hospital visit, or death.

Samsung Medical Center (SMC). We collected 209,523 patients who visited the SMC, were diagnosed at least once, and underwent blood tests between 2010 and 2012. We defined the baseline as 2010–2012. We excluded patients who were diagnosed with the target disease, and patients who died before January 1, 2013. We followed the eligible patients from January 1st, 2013, and the last follow-up date was the earliest date among the dates of the event, the last hospital visit (December 31st, 2020), or death.

2.5. Statistical analysis

We performed a Student's t-test and Pearson's correlation test to compare the correlations between the three biological factors and cosine similarity. We also measured the hazard risk of the target disease with individual risk scores using Cox proportional hazard regression. All statistical analyses were performed using R statistical software (version 3.6.1; R Foundation for Statistical Computing, Vienna, Austria). Statistical significance was declared at P-value < 0.05.

3. Results

We represented our main results with ICD2Vec based on GatorTron-OG, considering two aspects. First, GatorTron was pre-trained larger

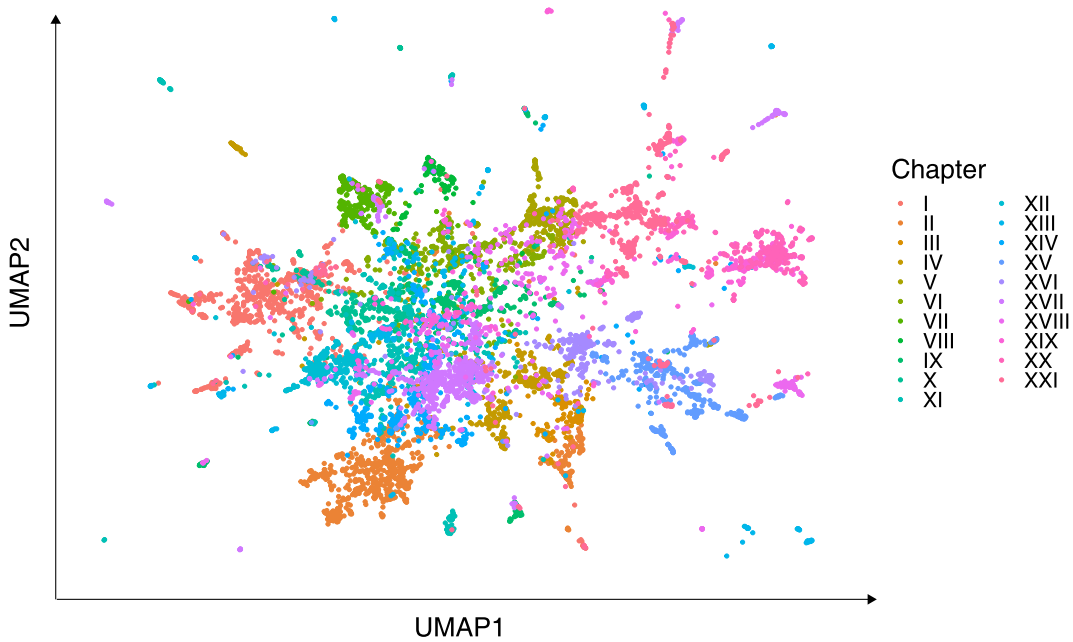

Fig. 2. Visualization for ICD2Vec. The 1,024 vector dimensions for ICD2Vec$_{\text{GatorTron-OG}}$ were reduced to two vector dimensions. The dots were colored according to the chapters of ICD-10-CM.

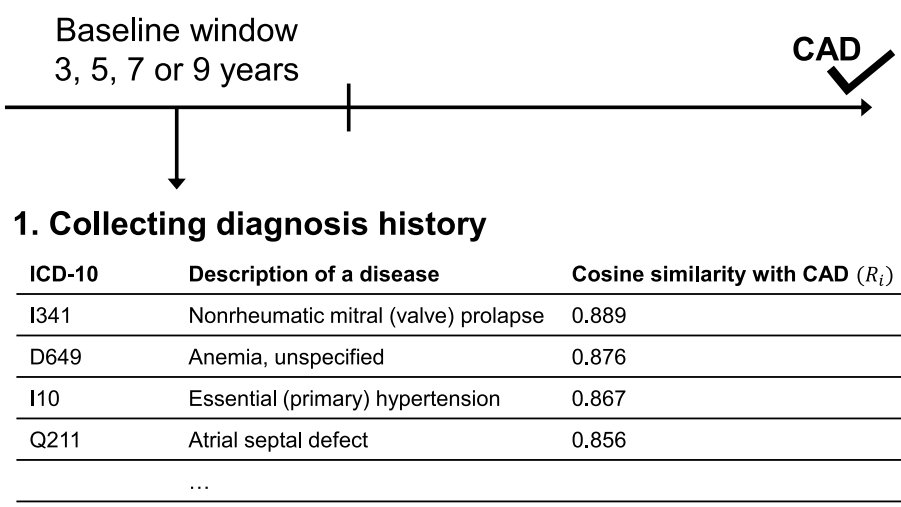

Fig. 3. Example of an epidemiological study design for identifying the relationship between IRIS$_{\text{CAD}}$ and CAD. A cohort study is designed to investigate the relationship between the IRIS and CAD. The time window of the baseline is defined as 3, 5, 7, or 9 years (only 3 years with SMC data). The disease-wide risk score (R_i) is defined as the cosine similarity between the target disease and other diseases. The maximum (R_{max}) or top three average (R_{avg3}) of the cosine similarities are defined as the IRIS in a patient diagnosed with several diseases at the baseline. Finally, the study participants are grouped into a decile based on the IRIS values. Abbreviations: the international classification of diseases (ICD); ICD2Vec risk score (IRIS); coronary artery disease (CAD); Samsung medical center (SMC).

biomedical corpus including medical notes than BioBERT or Bio + Clinical BERT, which can prevent a bias from a skewed word distribution, for example, between 222 words for A00 and 4 words for A01. Second, it is fine-tuned with five clinical NLP tasks more than others, it showed high performance in the previous work[32] by Yang et al. The results of other ICD2Vec models are described in **Supplementary Information**.

3.1. Arithmetic and semantic relationships between vectorized ICD codes

We empirically investigated the relationships between a symptom and a disease or between diseases using the analogical reasoning task shown by Mikolov et al. [22,38]. The task was performed by finding and evaluating the ICD code closest to the composite vector derived from the given vectors for the symptoms or diseases (Table 2). For example, the ICD code most similar to nearsightedness appears to be visual disturbances, refraction, or blindness and low vision (ICD-10: H53, H52, and H54). The contextualized vector derived from a description of itchy skin matches with pruritus, atopic dermatitis, other disorders of skin, and subcutaneous tissue (ICD-10: L29, L20, and L98). It is possible to use the calculation of diseases (e.g., I25 + I10) to find clinical manifestations with similar medical descriptions (e.g., I51). We created a vector for coronavirus disease 2019 (COVID-19), as summarized by the Mayo Clinic [39]. Interestingly, the diseases most similar to COVID-19 are J00, acute nasopharyngitis as a common cold; A99, unspecified viral hemorrhagic fever; and B03, smallpox. Relevant examples with ICD2Vec based on other BERT models or FastText are presented in **Supplementary Table S1**.

R00-R99 of ICD-10 indicates unclassified symptoms, signs, and abnormal clinical findings. We found the most similar ICD-10 code using an additive operation of up to three randomly selected R codes, using cosine similarity. This random selection was repeated 1,000 times. Table 3 lists the top 10 results according to the cosine similarity. For example, a combination of R11 (abnormal involuntary movements) and R56.0 (febrile convulsions) is indicative of G72.3 (periodic paralysis) with a cosine similarity of 0.97. A single symptom, R51 (Headache), is found to be a symptom of G43.9 (Migraine, unspecified).

Table 2
Examples of analogical reasoning with ICD2Vec based on GatorTron. The top three similar International Classification of Diseases (ICD)-10 codes were found based on the cosine similarity with a given vector for a symptom or disease. Semantic compositionality through vector operations such as addition was also possible when representing an ICD-10 code. We showed only ICD-10 codes as a category level (three-character code). Python codes are provided to check for the most similar ICD-10 code for the given symptoms or diseases (see Code availability).

Symptoms	Top three similar ICD-10 (definition of the disease) (cosine similarity)
Nearsightedness is a common vision condition in which you can see objects near to you clearly, but objects farther away are blurry.[55]	H53 (Visual disturbances) (0.93) H52 (Disorders of refraction and accommodation) (0.92) H54 (Blindness and low vision) (0.89)
Itchy skin is an uncomfortable, irritating sensation that makes you want to scratch.[56]	L29 (Pruritus) (0.91) L20 (Atopic dermatitis) (0.90) L98 (Other disorders of skin and subcutaneous tissue, not elsewhere classified) (0.89)
Coma is a state of prolonged unconsciousness that can be caused by a variety of problems.[57]	R40 (Somnolence, stupor and coma) (0.92) G93 (Other disorders of brain) (0.90) R57 (Shock, not elsewhere classified) (0.89)
Asthma is a condition in which your airways narrow and swell and may produce extra mucus.[58]	J45 (Asthma) (0.92) R06 (Dyspnoea) (0.91) J98 (Other respiratory disorders) (0.90)
Diseases	**Top three similar ICD-10**
I25 (Chronic ischemic heart disease) + I10 (Essential hypertension)	I51 (Complications and ill-defined descriptions of heart disease) (0.96) I20 (Angina pectoris) (0.95) G93 (Other disorders of brain) (0.95)
H34 (Retinal vascular occlusions) + E11 (Non-insulin-dependent diabetes mellitus)	E10 (Insulin-dependent diabetes mellitus) (0.95) H35 (Other retinal disorders) (0.95) E13 (Other specified diabetes mellitus) (0.94)
COVID-19 [39]	J00 (Acute nasopharyngitis [common cold]) (0.94) A99 (Unspecified viral hemorrhagic fever) (0.94) B03 (Smallpox) (0.94)

Abbreviations: the international classification of diseases (ICD).

Table 3
Examples of analogical reasoning with combinations of R codes. We randomly chose a combination of R codes and found the most similar ICD-10 code to the combination. The top 10 examples with the highest cosine similarity between the randomly chosen R codes and selected ICD-10 code are shown. We provided the code for constructing the full table (see Code availability).

R code 1	R code 2	R code 3	Most similar ICD-10 (definition of the disease) (CS)
R09.0 (Asphyxia)			T71 (Asphyxiation) (1.00)
R78.0 (Finding of alcohol in blood)			Y90.9 (Presence of alcohol in blood, level not specified) (0.98)
R41.3 (Other amnesia)			G45.4 (Transient global amnesia) (0.97)
R51 (Headache)			G43.9 (Migraine, unspecified) (0.97)
R11 (Abnormal involuntary movements)	R56.0 (Febrile convulsions)		G72.3 (Periodic paralysis) (0.97)
R41.2 (Retrograde amnesia)			G45.4 (Transient global amnesia) (0.97)
R23 (Other skin changes)			L98 (Other disorders of skin and subcutaneous tissue, not elsewhere classified) (0.97)
R17 (Unspecified jaundice)	R21 (Rash and other nonspecific skin eruption)	R51 (Headache)	T14.8 (Other injury of unspecified body region) (0.97)
R25.1 (Tremor, unspecified)			G24.9 (Dystonia, unspecified) (0.97)
R50.9 (Fever, unspecified)	R58 (Hemorrhage, not elsewhere classified)		T14.8 (Other injury of unspecified body region) (0.98)

Abbreviation: cosine similarity (CS); the international classification of diseases (ICD).

3.2. Comparison of ICD2Vec pairs with other disease–disease relationships

ICD2Vec was constructed based on the clinical words, sentences, and descriptions related to each disease. We sought to confirm the rationale and validity of ICD2Vec by comparing the biological relationships between diseases. We identified the significant associations between the cosine similarities obtained in our disease-to-disease pairs, derived from $\text{ICD2Vec}_{\text{GatorTron-OG}}$, and three biological indicators, i.e., the relative risk for comorbidities in observational clinical data, separation in a disease-disease network, and genetic correlations based on genome-wide association studies (**Supplementary Information**). The results from Med2Vec [19], ICD-Codex [34], and ICD2Vec based on other algorithms are also shown in **Supplementary Figure S1**. Our ICD2Vec algorithms generally showed higher correlations with biological relationships than Med2Vec and ICD-Codex (Table 4).

3.2.1. Relative risk of comorbidity

The relative risk was quantified as the ratio of co-occurrence of the two diseases to the incidence of each disease. We only considered statistically significant relative risks. A total of 127,579 selected disease pairs are presented in Fig. 4**a**. The correlation of the top 10% of the cosine similarity group was higher than that of the total set of pairs (0.281 vs. 0.195). There was a tendency for a higher relative risk between two diseases with a higher cosine similarity. The mean difference in the relative risk between disease pairs with the top 10% of cosine

Table 4
Pearson's correlations between cosine similarities and biological relationship with Med2Vec, ICD-Codex, and ICD2Vec pairs based on TF-IDF, FastText, BERT, fine-tuned BERT, fine-tuned BioBERT, fine-tuned Bio + Clinical BERT, and GatorTron-OG. The results from $ICD2Vec_{GatorTron\text{-}OG}$ are identical to the results in Fig. 2. The illustrations of the correlation plots are depicted in Supplementary Figure S1. Abbreviation: term frequency-inverse document frequency (TF-IDF).

Base model	Relative risk	Separation	Genetic correlation
Med2Vec [28] (200 dimensions)	0.091 (0.085 to 0.096)	–0.027 (–0.038 to –0.016)	–0.031 (–0.082 to 0.020)
ICD-Codex [60] (128 dimensions)	0.004 (–0.003 to 0.010)	–0.164 (–0.174 to –0.154)	0.023 (–0.027 to 0.074)
$ICD2Vec_{TF\text{-}IDF}$ (16,384 dimensions)	0.345 (0.340 to 0.349)	–0.203 (–0.213 to –0.193)	0.177 (0.128 to 0.225)
$ICD2Vec_{FastText}$ (300 dimensions)	0.117 (0.112 to 0.122)	–0.142 (–0.153 to –0.132)	0.080 (0.030 to 0.129)
$ICD2Vec_{BERT}$ (768 dimensions)	0.072 (0.067 to 0.078)	–0.171 (–0.182 to –0.161)	–0.022 (–0.073 to 0.027)
$ICD2Vec_{BERT}$ (fine-tuned, 768 dimensions)	0.150 (0.144 to 0.155)	–0.103 (–0.113 to –0.092)	0.068 (0.018 to 0.118)
$ICD2Vec_{BioBERT}$ (fine-tuned, 1,024 dimensions)	0.095 (0.090 to 0.105)	–0.049 (–0.060 to –0.038)	0.087 (0.037 to 0.136)
$ICD2Vec_{Bio+Clinical\ BERT}$ (fine-tuned, 768 dimensions)	0.111 (0.106 to 0.117)	–0.065 (–0.076 to –0.055)	0.092 (0.042 to 0.142)
$ICD2Vec_{GatorTron\text{-}OG}$ (fine-tuned, 1,024 dimensions)	0.195 (0.190 to 0.201)	–0.088 (–0.099 to –0.077)	0.232 (0.184 to 0.279)

similarity and the other group was statistically significant (P-value = 1.5×10^{-105}, Fig. 4**d**).

3.2.2. Separation in a disease interactome network

A previous study [21] by Menche et al. developed a network-based separation of two diseases as a measure of how close the diseases were in a disease–disease interactome network. In other words, the smaller the value, the more similar the diseases. We considered a total of 33,332 pairs with separation values under a *q*-value of 0.05. The correlation of the top 10% of the cosine similarity group was negatively higher than that of the total set of pairs (-0.240 vs. -0.088) (Fig. 4**b**). The mean difference of the top 10% group was statistically lower than that of the other groups (P-value = 7.9×10^{-112}, Fig. 4**e**).

3.2.3. Genetic correlation based on genome-wide association studies

A genetic correlation is a biological indicator that can be used to quantify disease-disease relationships. We compared the ICD2Vec cosine similarities with the genetic correlations of 1,531 pairs, as obtained using linkage disequilibrium score regression based on statistics of diseases from a genome-wide association study. There was a positive correlation between the genetic correlation and cosine similarity (Fig. 4**c**). The genetic correlations in the top 10% cosine similarity group were significantly higher than those in the other group (P-value = 8.6×10^{-15}, Fig. 4**f**).

3.3. Use-cases: epidemiological studies using IRIS

The association between IRIS and the risks for eight non-communicable diseases (namely, coronary artery disease (CAD), type 2 diabetes (T2D), dementia, liver cancer, cerebrovascular disease, chronic obstructive pulmonary disease (COPD), lung cancer, and chronic kidney disease (CKD)), which are the leading causes of death globally, was examined [40]. We designed prospective and retrospective cohort studies for each disease using data collected in the UK Biobank and with data extracted from the CDW of the SMC in South Korea. The incidence of eight diseases was defined using ICD-10 and ICD-9 (Fig. 3). The patients were grouped into deciles or 50-tiles according to the IRIS calculated for each patient.

We estimated the hazard ratios (HRs) and area under the receiver operating characteristics (AUROC) for the IRIS values using Cox proportional hazard models. The models were adjusted for age and sex. Additionally, we estimated HRs for baselines of 3, 5, 7, and 9 years in the UK Biobank and for a baseline of 3 years in SMC. Table 5 shows the demographic characteristics and highest HRs for each target disease with the two independent datasets. For example, in the analysis for CAD using 9 years for the baseline and IRIS using the top three averages in the UK Biobank, 3,735 patients were diagnosed with CAD in the top 10% of the $IRIS_{CAD}$ values, whereas 1,542 patients were diagnosed in the bottom 10% of the $IRIS_{CAD}$ values. The effect of the $IRIS_{CAD}$ was significantly associated with the risk of CAD (adjusted HR: 2.15, 95% confidence interval [CI] [2.02–2.28]; AUROC: 0.587, 95% CI [0.583 – 0.591]). The highest HR for each disease was selected among eight HRs obtained from the two methods (and) for four baseline years (**Supplementary Tables S3–S10**). Patients with a higher $IRIS_{CAD}$ tended to have a higher hazard for CAD (**Supplementary Figure S2**). In the SMC, with 3 years as the baseline and the top-three average of the $IRIS_{CAD}$ values, 1,471 patients were diagnosed with CAD in the top 10% of the $IRIS_{CAD}$ values, whereas 432 patients were diagnosed in the bottom 10% of $IRIS_{CAD}$ values (Table 5). The risk of CAD increased in SMC patients with higher $IRIS_{CAD}$ values (adjusted HR: 2.30, 95% CI [2.13–2.66]; AUROC: 0.612 95% CI [0.605 – 0.618]). Similarly, the HRs for other target diseases also increased in individuals with higher $IRIS_{CAD}$ values (**Supplementary Table S11** and **Supplementary Figure S3**). Moreover, a higher incidence of CAD was observed in groups with high scores binned according to the 50-tiles of the $IRIS_{CAD}$ in the UK Biobank (Fig. 5). For instance, the incidence of CAD in the highest risk score group was approximately three times higher than that in the lowest risk score group (18.3% vs. 6.3%). We further confirmed the performances of each model for the incidence of CAD (Table 6). The performances of ICD2Vec models are relatively higher than those of Med2Vec, ICD-Codex, and TF-IDF. The AUROC of GatorTron was significantly higher than that of TF-IDF (Delong's test, P-value $< 2.2 \times 10^{-16}$).

3.4. $IRIS_{CAD}$ for a common disease and comorbid conditions

The IRIS, calculated based on each individual's disease history, not only identifies the risk groups for a single target disease but also provides insights into comorbid conditions. To identify the comorbid conditions in a risk group based on the disease-wide risk score, we divided the study population at a baseline of 9 years into decile risk groups, as derived from the top three averages based on the onset of CAD. We used the 10 most common comorbidities for CAD and three other cardiovascular-related diseases as reported by the United States Department of Health & Human Services for comparison with actual prevalence [41]. The 13 comorbidities for CAD generally showed a higher prevalence among the high CAD risk groups (Fig. 6**a**). Of the 13 comorbidities, the prevalence of hypertension had the strongest association with the high CAD risk group 0.03% and 28.58% for the bottom 10% and top 10% CAD risk groups, respectively. In contrast, dementia had the lowest prevalence irrespective of the CAD risk group, with values of < 0.01% and 0.36% for the bottom 10% and top 10% CAD risk groups, respectively. In addition, the prevalence of other comorbidities similarly increased for the top 10% CAD risk group relative to the bottom 10% group, including hyperlipidemia (<0.01–10.62%), anemia (<0.01–7.44%), T2D (0.12–6.28%), atrial fibrillation (0.04–4.34%), COPD (0.01–3.91%), cataracts (1.54–9.40%), depression

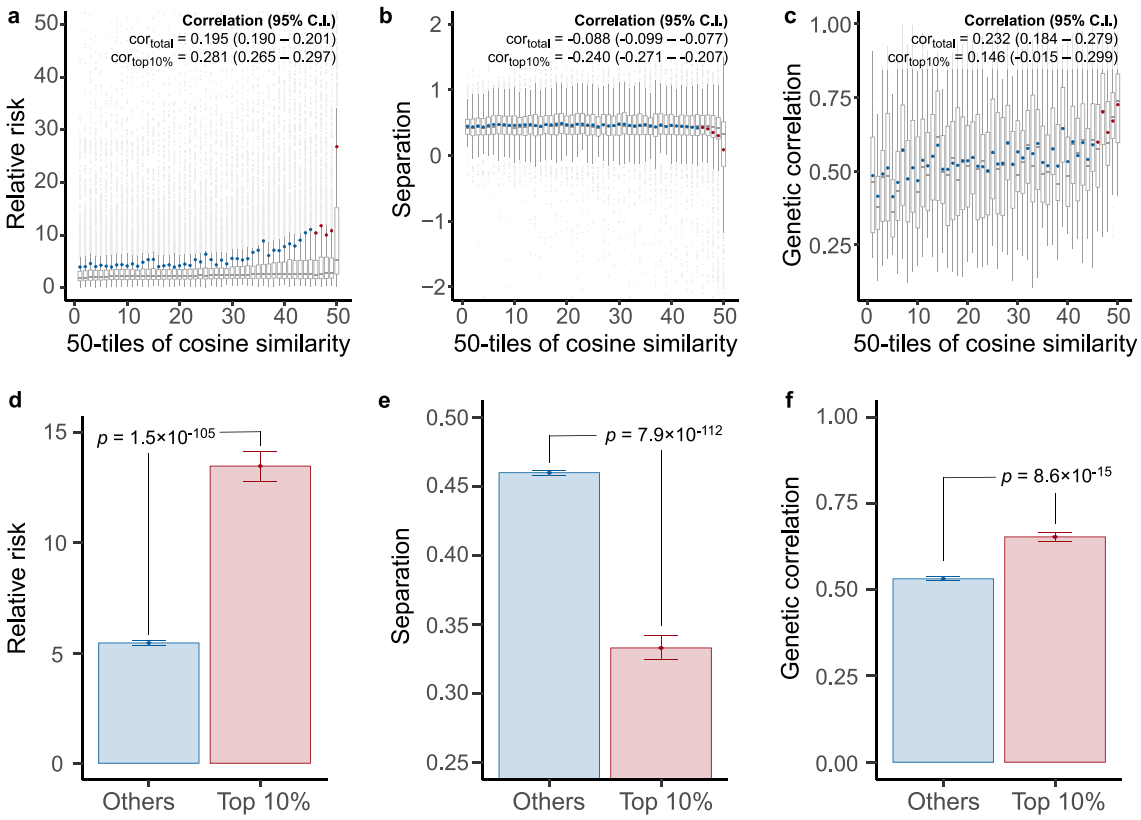

Fig. 4. Comparison between cosine similarities of ICD2Vec pairs and biological relationships between disease pairs. We defined three factors to measure biological relationships between disease pairs: relative risk (a, d), separation (b, e), and genetic correlation (c, f). The factors were compared with the cosine similarity between ICD2Vec pairs. The red dots and bars represent the average of the pairs with the top 10% of cosine similarity, and blue dots and bars represent that of the other 90% of the pairs. We measured the correlations between the cosine similarities from the total and top 10% ICD2Vec pairs and the biological relationships from corresponding disease pairs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(<0.01–6.70%), stroke (<0.01–2.70%), heart failure (<0.01–1.89%), rheumatoid arthritis (0.03–1.61%), and CKD (0.01–1.31%). The 20 most prevalent diseases in the highest CAD risk group are described in **Supplementary Table S12**. Either or without events, participants were most frequently diagnosed with hypertension (I10).

We further investigated the risk of CAD among subgroups as divided by the $IRIS_{CAD}$ and 10-year atherosclerotic cardiovascular disease (ASCVD) risk. The 10-year risk for ASCVD was categorized according to the definition by the American College of Cardiology Foundation [42]. A high $IRIS_{CAD}$ was associated with CAD risk, even within the same 10-year ASCVD risk group (Table 7). The cumulative CAD incidence of all other groups was higher than that of the 1st decile of $IRIS_{CAD}$ values with a low 10-year ASCVD risk (Fig. 6**b**). In particular, the top 10% of $IRIS_{CAD}$ values and high 10-year ASCVD risk groups showed substantially increased CAD risks (adjusted HR, 4.26; 95% CI, 3.59–5.05).

4. Discussion

ICD-10 is a global standardized hierarchical code for representing diseases. In addition to its medical purposes, such as in clinical diagnosis, ICD-10 is essential in biomedical research because it is used to define disease variables [8]. In conventional analyses, such qualitative data are represented as dummy variables or one-hot encoding variables randomly distributed in the latent space; however, this representation loses the intrinsic relationships with other related variables. Accordingly, we proposed ICD codes that were vectorized to contain meaningful measurable quantitative values and suggested a mathematical representation of diseases using ICD-10, called ICD2Vec. We showed that ICD2Vec contained the semantic and arithmetic relationships among diseases and presented medically conceptual meanings (Table 2). Moreover, the similarity of the vectors was significantly correlated with known biological relationships (Fig. 4). Furthermore, we found a significant association between the individual risk score derived from ICD2Vec and the actual disease incidence based on prospective cohort analyses of eight diseases across two independent datasets from the UK and South Korea (Table 5 and Fig. 5), as discussed above. The higher risk score group for CAD (high $IRIS_{CAD}$ carriers) had more CAD-related comorbid conditions (Fig. 6 and Table 7).

Previous encoding techniques have used converted vectors to provide information [22,36,43]. These techniques have also been applied to a variety of concepts in medicine and biology [18,44]. In some studies, mathematical vectors embedding medical histories were used to predict the future status of patients [45,46]. Applications to unstructured medical records provided novel insights for quantifying the relationships between diseases in multi-dimensional vector spaces [45]. However, these algorithms have limitations; the resulting vectors are dependent on the reference cohort. It arises a bias to shift the distribution of a disease according to patient's physical status or environment. In contrast, ICD2Vec is universal ICD vectors to be applicable to any cohort. This cohort-free framework is simple to convert an ICD code to the corresponding vector. Previous algorithms showed improved clinical NLP tasks. However, they were not used to convert ICD codes to vectors, although it is reasonable and useful to use vectorized ICD codes to analyze medical conditions. A study which was conducted to elucidate the relationships between embedded vectors and clinical definitions compared their own vectors with medical concepts rated by physicians [47]. Their vectors were validated as embedded vectors for readmission prediction based on a large cohort set. We quantitatively demonstrated that the vectors of ICD2Vec were meaningfully distributed in the latent space through a statistical investigation of the relationship between

Table 5
Highest hazard ratios of ICD2Vec risk score for eight diseases.

Dataset	Target disease	Baseline (risk score derivation)	Top 10% of IRIS, [no. of samples, (no. of events)]	Bottom 10% of IRIS [no. of samples, (no. of events)]	Hazard ratio* (95% CI)	AUROC (95% CI)
UK Biobank	CAD	9 years (top three average)	25,774 (3,735)	25,774 (1,542)	2.15 (2.02 – 2.28)	0.587 (0.583 – 0.591)
	T2D	9 years (top three average)	26,826 (2,923)	26,827 (1,203)	2.28 (2.14 – 2.44)	0.579 (0.575 – 0.583)
	Dementia	9 years (maximum)	27,737 (1,099)	27,738 (246)	2.85 (2.48 – 3.27)	0.623 (0.615 – 0.631)
	Liver cancer	7 years (maximum)	24,683 (96)	24,684 (20)	4.51 (2.78 – 7.30)	0.599 (0.577 – 0.621)
	Cerebrovascular diseases	9 years (top three average)	27,349 (2,329)	27,350 (671)	2.42 (2.22 – 2.64)	0.609 (0.604 – 0.614)
	COPD	9 years (maximum)	27,482 (3,302)	27,483 (612)	4.67 (4.28 – 5.09)	0.657 (0.652 – 0.662)
	Lung cancer	9 years (maximum)	27,727 (652)	27,728 (222)	2.23 (1.92 – 2.60)	0.590 (0.581 – 0.600)
	CKD	9 years (top three average)	27,682 (4,036)	27,683 (617)	4.81 (4.42 – 5.24)	0.677 (0.672 – 0.682)
SMC	CAD	3 years (top three average)	13,664 (1,471)	13,665 (432)	2.30 (2.06 – 2.57)	0.612 (0.605 – 0.618)
	T2D	3 years (maximum)	13,036 (2,334)	13,037 (450)	5.14 (4.64 – 5.69)	0.635 (0.629 – 0.642)
	Dementia	3 years (top three average)	14,737 (1,116)	14,738 (113)	3.21 (2.63 – 3.91)	0.673 (0.665 – 0.682)
	Liver cancer	3 years (maximum)	14,790 (599)	14,790 (187)	3.40 (2.89 – 4.01)	0.549 (0.540 – 0.557)
	Cerebrovascular diseases	3 years (maximum)	13,507 (1,695)	13,508 (937)	1.27 (1.17 – 1.38)	0.562 (0.557 – 0.567)
	COPD	3 years (top three average)	14,899 (457)	14,900 (86)	3.58 (2.83 – 4.52)	0.643 (0.630 – 0.656)
	Lung cancer	3 years (maximum)	14,929 (267)	14,930 (98)	1.51 (1.19 – 1.91)	0.586 (0.571 – 0.602)
	CKD	3 years (maximum)	14,592 (1,126)	14,593 (102)	7.78 (6.34 – 9.54)	0.693 (0.686 – 0.701)

Hazard ratio was estimated using a Cox proportional hazard model adjusted for sex and age.
*All P-values for hazard ratios were estimated under 2.2×10^{-16}.
Abbreviations: Samsung Medical Center (SMC); coronary artery disease (CAD); type 2 diabetes (T2D); the international classification of diseases (ICD); ICD2Vec risk score (IRIS); confidence interval (CI); standard deviation (SD).

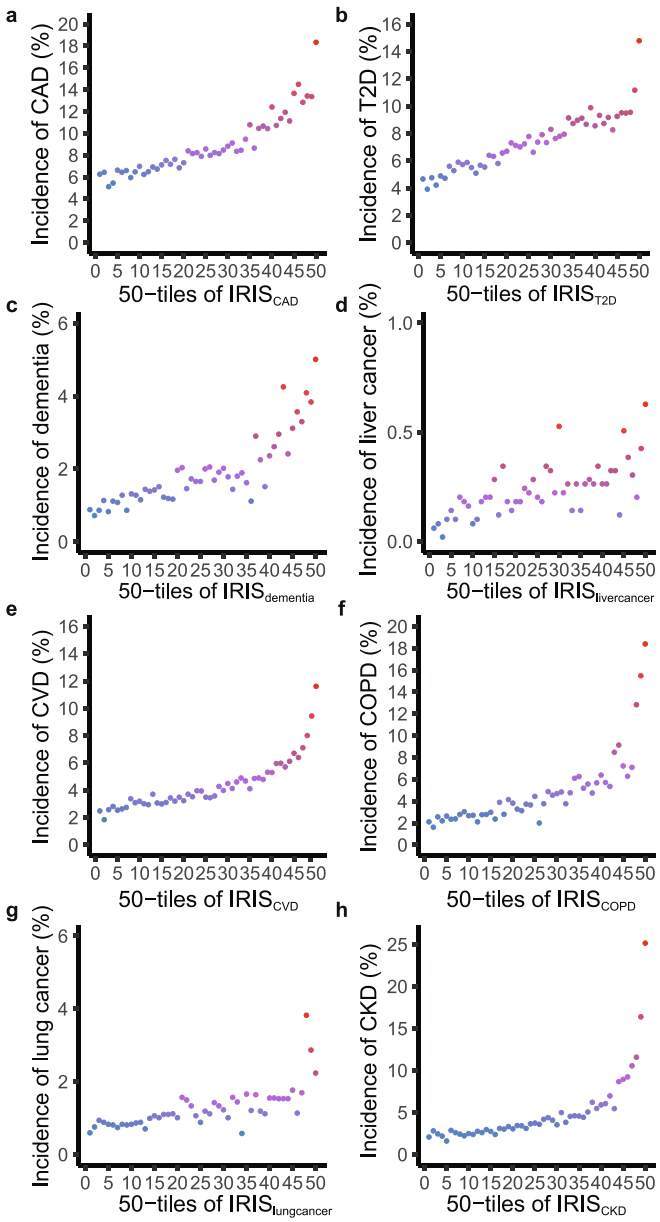

Fig. 5. Incidence of four diseases according to IRIS risk groups (UK Biobank). (a) CAD. (b) T2D. (c) dementia. (d) liver cancer. (e) CVD. (f) COPD. (g) lung cancer. (h) CKD. Abbreviations: the international classification of diseases (ICD); ICD2Vec risk score (IRIS); coronary artery disease (CAD); type 2 diabetes (T2D); cerebrovascular disease (CVD); chronic obstructive pulmonary disease (COPD); chronic kidney disease (CKD).

disease pairs based on three biologically established indicators (comorbidity as a relative risk, separation, and genetic correlation). These results suggest that ICD2Vec using advanced biologically fine-tined and/or clinically fine-tuned BERT and GatorTron-OG can provide improved performance and superior biological interpretation compared to existing ICD-Codex (node2Vec-based) and Med2Vec (word2Vec-based) methods which are based on relatively simple NLP and embedding concepts trained with cohort data. ICD2Vec, different from other algorithms, can estimate the most similar disease for the unknown disease. We can infer the similar disease for an unknown disease such as COVID-19 even not classified in ICD-10.

We also confirmed the utility of ICD2Vec across the association study using the UK Biobank and the SMC cohort. In addition, we defined individual risk scores based on ICD2Vec and performed use-case studies to investigate whether the scores for past diseases were effective in

Table 6
Results for CAD risk according to baseline and IRIS derivation using the UK Biobank.

Base model	Top 10% of IRIS [N (events)]	Bottom 10% of IRIS [N (events)]	Hazard ratio** (95% CI)	AUROC (95% CI)
Med2Vec [28] (200 dimensions)	25,774 (1,006)	25,774 (1,670)	0.75 (0.69 – 0.81)	0.471 (0.467 – 0.476)
ICD-Codex [60] (128 dimensions)	25,774 (2,742)	25,774 (1,976)	1.54 (1.46 – 1.64)	0.537 (0.533 – 0.541)
ICD2Vec$_{TF\text{-}IDF}$ (16,384 dimensions)	25,774 (3,385)	25,774 (1,641)	2.26 (2.13 – 2.40)	0.569 (0.565 – 0.573)
ICD2Vec$_{FastText}$ (300 dimensions)	25,774 (4,190)	25,774 (1,527)	2.24 (2.11 – 2.37)	0.591 (0.587 – 0.595)
ICD2Vec$_{BERT}$ (768 dimensions)	25,774 (3,608)	25,774 (1,786)	1.83 (1.73 – 1.94)	0.570 (0.566 – 0.574)
ICD2Vec$_{BERT}$ (fine-tuned, 768 dimensions)	25,774 (3,958)	25,774 (1,594)	2.21 (2.09 – 2.35)	0.582 (0.578 – 0.586)
ICD2Vec$_{BioBERT}$ (fine-tuned, 1,024 dimensions)	25,774 (3,832)	25,774 (1,700)	2.01 (1.90 – 2.13)	0.578 (0.574 – 0.582)
ICD2Vec$_{Bio+Clinical\ BERT}$ (fine-tuned, 768 dimensions)	25,774 (3,805)	25,774 (1,582)	2.19 (2.06 – 2.32)	0.583 (0.579 – 0.587)
ICD2Vec$_{GatorTron\text{-}OG}$ (fine-tuned, 1,024 dimensions)	25,774 (3,735)	25,774 (1,542)	2.15 (2.02 – 2.28)	0.587 (0.583 – 0.591)

*Hazard ratio was estimated using a Cox proportional hazard model adjusted for sex and age.
**All P-values for hazard ratios were estimated under 2.2×10^{-16}. Abbreviations: the international classification of diseases (ICD); ICD2Vec risk score (IRIS); coronary artery disease (CAD); standard deviation (SD); confidence interval (CI).

classifying risk groups. As a result, the risk of a target disease was predicted using these meaningful vectors. We observed a significant association between the risk of a target disease and the IRIS values for all other diseases during a specific baseline period (Table 5 and Fig. 5). In our experiments, the disease incidence of patients in the high-risk group was higher than that in the low-risk group (**Supplementary Tables S3–S10** and **Supplementary Figures S2–S3**). Specifically, we identified ICD codes related to the 10 most common comorbidities for CAD, which were previously reported as diseases with a significant association with CAD [41] (Fig. 6). Patients in the high-risk group were more likely to have underlying diseases relevant to the target disease over several years before the manifestation of the target disease. We confirmed that patients with a higher 10-year ASCVD risk and higher IRIS$_{CAD}$ risk group were likely to develop CAD (Table 7). Notably, the IRIS index derived from ICD2Vec is intuitive and simple to calculate.

ICD2Vec is publicly available and applicable to various fields. Several examples of possible applications have been suggested. First, the IRIS values of all other diseases corresponding to a target disease can be computed and used as a risk predictor or covariate. Typically, researchers define arbitrarily selected relevant diseases as covariates or define an index for scoring whether participants were diagnosed with other diseases using the Charlson Comorbidity Index. Although conventional approaches adjust the main effect restrictively with only diseases selected and known by researchers, the IRIS is computed from all diseases and can be used to analyze a disease with unknown covariates. Second, a multimodal analysis requires efficient vectors for the features

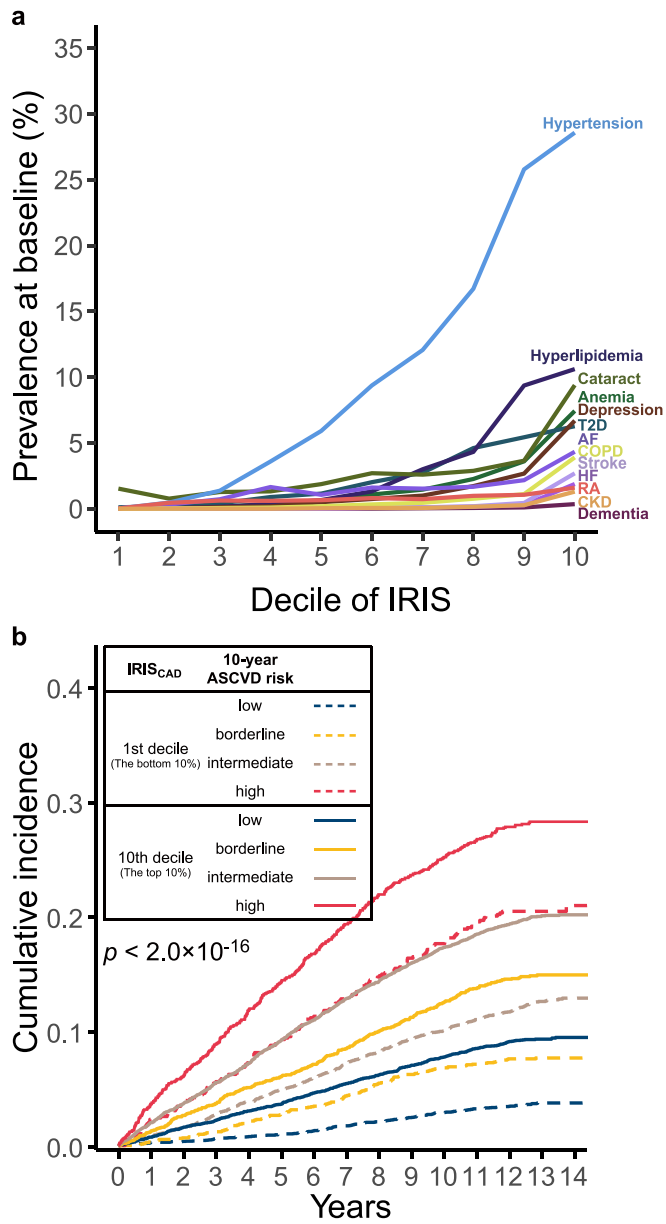

Fig. 6. (a) Prevalence of 13 comorbidities for CAD according to $IRIS_{CAD}$ risk groups. (b) Cumulative incidences of subgroups by $IRIS_{CAD}$ and 10-year ASCVD risk (the UK Biobank). Abbreviations: the international classification of diseases (ICD); ICD2Vec risk score (IRIS); type 2 diabetes (T2D); chronic kidney disease (CKD); heart failure (HF); rheumatoid arthritis (RA); atrial fibrillation (AF); chronic obstructive pulmonary disease (COPD); atherosclerotic cardiovascular disease (ASCVD).

of a disease. The size of ICD2Vec can be lower than those of one-hot encoded ICD-10 codes. The dimensions (1,024) of ICD2Vec based on GatorTron are reduced approximately 24 times relative to the original dimensions (24,354) of the ICD codes. Third, using pre-trained vectors, it is possible to increase the performance of a predictive model. Owing to the pre-trained knowledge embedded in ICD2Vec, the parameters of a model can be well optimized by ICD2Vec, i.e., better than the approach with randomly defined variables used by one-hot encoding. A recent review argues that the embedding of clinical concepts is important because of dimensionality reduction for better training [48]. Lastly, ICD2Vec might be helpful in mapping ICD standards to other standard terminologies, such as Systematized Nomenclature of Medicine - Clinical Terms (SNOMED-CT) and Medical Subject Headings. By calculating the cosine similarity between codes of different standards with an algorithm such as ICD2Vec, code mapping can be performed in the order of the highest similarity and relevance. We can further study the vector representation using SNOMED-CT. SNOMED-CT includes not only disease but also clinical operation, measurement, and concepts. Embedding clinical concepts with SNOMED-CT may be powerful because mapping SNOMED-CT to other standard codings like ICD-10 or PheCode [61] is essential. However, enough clinical information for each SNOMED-CT code is required to generate a meaningful vector. As we have shown, ICD2Vec can be adapted to other clinical concept standards if there is a mapping list. Instead, in the future, we can simply develop ICD2Vec for the other standards such as SNOMED-CT and ICD-11 using the pipeline.

Our study had several limitations. First, analogical reasoning with ICD2Vec does not represent a clinical diagnosis. Clinical diagnosis involves systematic clinical reasoning with patient information such as disease history, physical examination, and laboratory results [49–51]. For example, the key diagnostic factor of pulmonary edema is excessive fluid, as revealed by medical lung imaging tests [52]. Similar ICD codes are not the results of a diagnosis but have similar sources. Thus, in this study, we presented the top three similar ICD codes to illustrate the arithmetic and semantic relationships among the ICD2Vec codes. Second, converting ICD codes to vectors largely depends on the descriptions of the ICD codes and medical knowledge. Therefore, ICD2Vec needs to be updated as medical knowledge accumulates. Third, much larger-scale algorithms such as Generative Pre-trained Transformer-3 can be used to generate vectors [53]. In this study, we showed that the ICD2Vec approach performed better than other existing algorithms. However, the performance comparison among ICD2Vec's application models needs additional experiments. Therefore, the addition and comparison of various advanced NLP models are needed to investigate the best-fit model with ICD2Vec and to improve the performance of ICD2Vec in a further study. The relatively simple information for diseases (e.g., medical definitions and explanations for ICD codes) might be insufficient to develop well-established vectors, and thus, ICD2Vec models also need to be trained with additional knowledge-based biomedical data corresponding to each ICD code. A comparison of different algorithms is required to improve the performance of ICD2Vec. Fourth, models pre-trained with English Wikipedia and a large corpus may have a bias for gender and racial stereotypes [54]. The bias may affect ICD2Vec through certain medical conditions. Further research for evaluating the stereotypical associations is needed.

Table 7
Adjusted hazard ratio (95% CI) for CAD of subgroups by $IRIS_{CAD}$ and 10-year ASCVD risk.

		10-year ASCVD risk			
		Low	Borderline	Intermediate	High
$IRIS_{CAD}$	Bottom 10% (1st decile)	1 (reference)	1.61 (1.32–1.97)	2.11 (1.80–2.48)	2.92 (2.36–3.61)
	Intermediate (2nd-9th decile)	1.37 (1.20–1.57)	2.00 (1.74–2.30)	2.53 (2.20–2.92)	3.19 (2.74–3.72)
	Top 10% (10th decile)	2.61 (2.23–3.06)	3.32 (2.81–3.92)	3.55 (3.06–4.11)	4.26 (3.59–5.05)

Abbreviations: the international classification of diseases (ICD); ICD2Vec risk score (IRIS); coronary artery disease (CAD); atherosclerotic cardiovascular disease (ASCVD).
[a]Hazard ratio was estimated using a Cox proportional hazard model adjusted for sex and age.

ICD2Vec is a framework for converting qualitatively measured ICD codes into quantitative vectors containing the semantic relationships among diseases. The publicly available ICD2Vec can be used in diverse research and medical practices.

4.1. Data sharing

Our ICD2Vec embedding dataset is available at https://github.com/YeongChanLee/ICD2Vec. The clinical information of ICD-10-CM is denoted in https://medlineplus.gov/connect/application.html#diagnosis. We crawled the information on ICD-10 in https://www.icd10data.com. The information for disease modules like network-based separation within the interactome is available at supplementary materials in Menche et al [21]. Med2Vec for comparison with ICD2Vec is available in https://github.com/mp2893/med2vec and we obtained embedding matrix using Med2Vec in https://mp2893.com/experiments.html. ICD-Codex is available at https://icd-codex.readthedocs.io/en/latest/. The UK Biobank GWAS summary statistics are available at http://www.nealelab.is/uk-biobank. The genotype and phenotype data can be obtained from the UK Biobank (https://www.ukbiobank.ac.uk) upon project application. EHRs of SMC are used for permitted researchers.

Code to construct ICD2Vec is available at https://github.com/YeongChanLee/ICD2Vec. We obtained the pre-trained BERT models such as BERT, BioBERT, and Bio + Clinical BERT from Hugging Face (https://huggingface.co/models). The paths of the pre-trained BERT models in Hugging Face are as follows: "bert-base-uncased" for BERT, "dmis-lab/biobert-large-cased-v1.1" for BioBERT, and "emilyalsentzer/Bio_ClinicalBERT" for Bio + Clinical BERT.

5. Author's contributions

Y.C.L., S.-H.J., and H.-H.W. designed the study. Y.C.L., S.-H.J., A.K., W.-Y.P., and H.-H.W. curated the datasets for this study. I.S., M.S. and Y.C.L. conducted data preprocessing. Y.C.L., S.-H.J., and A.K. performed the deep learning analysis, statistical analysis, and external validation. Y.C.L., K.K., and H.-H.W. reviewed the statistical results. M.S.K. and W.M. clinically interpreted the results. Y.C.L., S.-H.J., and A.K. wrote a manuscript under a supervision of H.-H.W. All authors contributed critical revisions of the manuscript.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Woong-Yang Park reports a relationship with GENINUS that includes: employment.

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Appendix A. Supplementary material

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